local-1777895137923

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	**local-1777895137923**
Industry Domain	**hvac_commercial_heating**
Engine Version	**PDF Engine v2.1 (Full Detail)**
Target Quantity	**Not specified**

ARCHITECTURE AT A GLANCE

MODULES	BILL OF MATERIALS ROWS	IDENTIFIED RISKS
8	**33**	**24**

ECONOMICS

EST. UNIT COST	TARGET CEILING	HEADROOM
£6,495.00	**None**	**N/A**

TOTAL NON-RECURRING ENGINEERING	UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING
£4,800.00	**£11,295.00**

Contents

1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Space heating and low-to-medium temperature domestic hot water provision for small commercial premises seeking to transition from legacy fossil fuel boilers to sustainable electrification.

Mission Statement

To accelerate the decarbonization of UK small commercial premises by providing an affordable, ultra-compact, and highly efficient R290 air-source heat pump.

Target Customers

Small to medium enterprises (SMEs), retail chains, boutique hotels, small office buildings, and light industrial facilities in the UK.

Why Now (Market Timing)

Strict UK net-zero targets, tightening F-gas regulations phasing out high-GWP refrigerants, and rising grid energy costs are driving rapid electrification of heating.

Full Synthesis Report

The UK commercial heating sector is currently undergoing a massive shift towards decarbonization, driven by stringent government net-zero targets and phase-outs of traditional gas boilers. The proposed 30kW air-source heat pump represents a highly compelling offering for the small commercial building sector, including retail spaces, small offices, and light industrial units. By utilizing R290 (propane) as the refrigerant, this system achieves an ultra-low Global Warming Potential (GWP) of 3, aligning perfectly with incoming F-Gas regulations that strictly limit and phase down higher GWP alternatives like R410A or R32. Achieving a Coefficient of Performance (COP) of 3.5 at A2/W35 is technically robust and meets the performance expectations for modern high-efficiency setups qualifying for UK incentives. However, housing a full 30kW R290 system within the tight dimensional constraints of 1200mm W x 800mm D x 1400mm H, while maintaining an aggressive noise profile below 65dB at 1m, poses a significant thermal and acoustic engineering challenge. The use of a scroll compressor and axial fan must be carefully paired with advanced acoustic enclosures and vibration dampening to meet these noise limits. Furthermore, the aggressive target unit cost of £4,500 at low-medium manufacturing volumes (500 units per year) will require rigorous supply chain optimization. Procuring the commercial-grade stainless steel heat exchanger and R290-rated spark-proof components at competitive price points will be the primary barrier to margin goals. Achieving domestic MCS certification is also critical for market viability and government subsidies, which necessitates strict adherence to both thermal performance standards and stringent safety standards for A3 highly flammable refrigerants.

1.2 Market & Competitors

Competitor Landscape

Mitsubishi Electric

Product	**Ecodan CAHV / QAHV Commercial Range**
Pricing Strategy	**£8,500 - £12,000 per unit**
Technical Specs	**40kW output, R407C or CO2 refrigerant, robust scroll compressor, high temperature capability.**

STRENGTHS

Deep UK distribution network, exceptionally reliable, strong brand trust.

WEAKNESSES

Much larger footprint than 1200x800mm, capital cost is significantly above our £4.5k target.

DIFFERENTIATION ANGLE

Our product targets a more compact, sub-£5k price point with low-GWP R290 specifically optimized for the 30kW small commercial niche.

Vaillant

Product	**aroTHERM plus (Cascaded setup)**
Pricing Strategy	**£3,500 per 15kW unit (Approx. £7,000 total for a 30kW cascaded loop)**
Technical Specs	**R290 refrigerant, max 15kW individual capacity, COP ~4.0, ultra-quiet.**

STRENGTHS

Exceptional acoustic design, natural R290 refrigerant, high thermal efficiency.

WEAKNESSES

Does not offer a native 30kW single module; requires cascading two units which expands installation footprint and complexity.

DIFFERENTIATION ANGLE

Providing a single, monolithic 30kW unit simplifies small commercial installation plumbing and halves the outdoor footprint compared to a dual-domestic unit cascade.

Clade Engineering

Product	**Chestnut R290 Air Source Heat Pump**
Pricing Strategy	**£15,000+**
Technical Specs	**Bespoke commercial build, 50kW+ capacity, R290 refrigerant, heavy duty industrial fans.**

STRENGTHS

UK-based bespoke engineering, incredibly robust multi-compressor industrial build, high temperature performance.

WEAKNESSES

Designed for medium-to-large commercial spaces; physically massive and prohibitively expensive for a small retail shop or office.

DIFFERENTIATION ANGLE

Perfectly addresses the missing 'light commercial' tier, capturing 30kW customers for whom Clade is too large/expensive, but conventional domestic brands lack single-unit thermal density.

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	**£0.00**
Max Mass	**-**
Target Process	
Target Material	
Tolerance Target	
Production Quantity	

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
UK F-Gas Regulation Updates	Government Policy	2026	Dictates the timeline for phasing out high-GWP refrigerants, validating the strategic choice of R290.
MCS Heat Pump Standard (MIS 3005-I)	Certification Standard	2026	Core compliance document for UK heat pump installations, mandatory for grant eligibility.
EN 378: Refrigerating systems and heat pumps	Safety Standard	2026	Mandatory safety and environmental requirements for designing systems with A3 highly flammable refrigerants.
Acoustics - Noise from heat pumps (BS EN 12102)	Engineering Standard	2026	Crucial testing framework for validating the stringent sub-65dB at 1m noise requirement.
CIBSE AM17: Heat pumps for commercial buildings	Design Guide	2026	Provides industry best practices for designing, sizing, and deploying commercial ASHPs in the UK market.

2.1 MCS MIS 3005-I

Microgeneration Certification Scheme

Status: not-started

Owner: Compliance Engineer

Standard Summary

Standard for the installation and product approval of heat pumps in the UK.

Applicability Assessment

> Overall product architecture and system installation.

Engineering Design Impact

Requires validated COP >= 3.5 at specified conditions, weather compensation algorithms, and strict documentation.

EVIDENCE REQUIRED

Lab test reports from a UKAS accredited facility proving performance data.

GAP ACTION

Identify an MCS-accredited test lab and book a preliminary performance evaluation.

2.2 BS EN 378-2

Safety and environmental requirements - Design and construction

Status: not-started

Owner: Mechanical Design Engineer

Standard Summary

Safety requirements for refrigerating systems containing flammable gases.

Applicability Assessment

> Indoor and outdoor unit design, specifically regarding R290 (A3) handling.

Engineering Design Impact

Requires spark-free electrical components (ATEX), proper ventilation paths in case of leaks, and strict charge limits.

EVIDENCE REQUIRED

Risk assessment, component ATEX certificates, and leak test simulation data.

GAP ACTION

Audit all electrical components near the compressor and heat exchanger for spark-free compliance.

2.3 BS EN 14511

Air conditioners, liquid chilling packages and heat pumps

Status: not-started

Owner: Thermal Engineer

Standard Summary

Testing and rating of heat pumps at standard conditions.

Applicability Assessment

> Thermal performance rating metrics.

Engineering Design Impact

Stainless steel heat exchanger and scroll compressor sizing must confidently hit the 30kW output and 3.5 COP baseline.

EVIDENCE REQUIRED

Standardized performance testing report under EN 14511 conditions.

GAP ACTION

Build digital thermal models to confirm 30kW target at standard A2/W35 and A7/W35 test points.

2.4 BS EN 12102-1

Determination of the sound power level

Status: not-started

Owner: Acoustics Engineer

Standard Summary

Acoustic emissions testing protocols for heat pumps.

Applicability Assessment

> Outdoor unit casing, fan, and compressor mount design.

Engineering Design Impact

Dictates the layout of acoustic insulation materials, axial fan speed profiles, and compressor isolation mounts.

EVIDENCE REQUIRED

Sound power level chamber test results displaying emissions below 65dB at 1m.

GAP ACTION

Design custom acoustic enclosure prototype and perform baseline sound frequency spectrum mapping.

2.5 CE/UKCA PED

Pressure Equipment Directive

Status: not-started

Owner: Manufacturing Engineer

Standard Summary

Safety standards for the design and fabrication of pressure vessels.

Applicability Assessment

> Refrigerant loop, stainless steel heat exchanger, and compressor body.

Engineering Design Impact

Piping and internal heat exchanger components must withstand the operational pressures of R290 with mandatory safety margins.

EVIDENCE REQUIRED

Pressure tests, welding certificates, and material traceability reports.

GAP ACTION

Source a PED-compliant stainless steel heat exchanger block and request preliminary supplier documentation.

3. Sizing & Spatial Optimisation

INFEASIBLE DESIGN ALLOCATION

The current sizing constraints are not satisfied. Module requirements exceed the available envelope.

System Envelope

KIND	RULES DOMAIN	FEASIBLE
generic_room	hvac_commercial_heating	No

Internal Dimensions (W×D×H)	**5000 × 5000 × 3000 mm**
Total Internal Floor Area	**25 m²**
Total Internal Volume	**75 m³**
Available Floor Budget	**21.25 m²**

Module Dimension Allocations

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALE
compressor	2236 × 2236 × 2400	5	floor	mass_proportional
evaporator	2236 × 2236 × 2400	5	floor	mass_proportional
fanmodule	2236 × 2236 × 2400	5	floor	mass_proportional
hydrocondenser	2236 × 2236 × 2400	5	floor	mass_proportional
refrigerantcircuit	2236 × 2236 × 2400	5	floor	mass_proportional
inverterdrive	2236 × 2236 × 2400	5	floor	mass_proportional
controlbox	2236 × 2236 × 2400	5	floor	mass_proportional

chassis	2236 × 2236 × 2400	5	floor	mass_proportional

Identified Conflicts

- Total module footprint (40.0m2) exceeds envelope budget (21.3m2)

Optimisation Recommendations

- Consider a larger envelope
- Optimize module footprint

4.1 R290 Scroll Compressor Assembly

Lead Time: 12 weeks | Status: draft

Module Purpose

Compresses low-pressure R290 vapor into high-pressure, high-temperature vapor to drive the thermal cycle.

Why it matters to the system

> It is the primary mover of the refrigerant loop, dictating both the maximum thermal output (30kW) and the bulk of the baseline noise emissions.

Detailed Technical Description

The foundation of the thermal engine. This module takes low-pressure R290 vapor and utilizes an inverter-driven scroll mechanism to compress it to high pressure and temperature. Due to R290's specific thermodynamic properties and solubility, the compressor requires specialized POE oil lubrication and heavy-duty seals to prevent long-term degradation.

The acoustic and vibration constraints for this 30kW unit are structurally paramount. The compressor is housed in a dense mass-loaded vinyl and lead-lined acoustic jacket to attenuate high-frequency mechanical noise, keeping the system compliant with the strict <65dB limit. It sits on tuned elastomeric isolators to prevent ground-borne vibration from turning the chassis into a resonant speaker.

SYSTEM INPUTS

- Low-pressure R290 vapor
- Variable 3-phase AC power

SYSTEM OUTPUTS

- High-pressure R290 vapor

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4.1.2 Specifications & Risk Profile

Expected Key Parts

- Inverter scroll compressor
- Acoustic compressor jacket
- Anti-vibration elastomeric mounts
- Crankcase heater

Identified Failure Modes

- Scroll bearing wear due to poor lubrication return at low loads
- Crankcase heater failure leading to liquid slugging on startup
- Acoustic jacket trapping excessive heat causing thermal overload

Open Questions

- Exact vibration transmissibility profile to the base pan
- Optimal lowest inverter frequency permissible without causing oil starvation

4.1.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
compressor-001	Inverter scroll compressor	Purchased	-	£800.00
compressor-002	Acoustic compressor jacket	Purchased	-	£800.00
compressor-003	Anti-vibration elastomeric mounts	Purchased	-	£40.00
compressor-004	Crankcase heater	Purchased	-	£40.00
			Module Total Cost:	**£1,680.00**

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4.1.4 Engineering Review

Review by Fang

FAIL

Verdict Summary

The design fails manufacturing compliance due to the specification of a lead-lined jacket, violating RoHS and introducing severe EH&S risks, while the acoustic enclosure creates dangerous R290 gas and heat trapping hazards.

Identified Issues

MATERIALS

The specification of a 'lead-lined' acoustic jacket is a direct violation of RoHS compliance and poses massive Environmental Health and Safety (EH&S) risks during assembly, handling, and disposal.

Suggestion: Replace lead with barium sulfate-loaded mass-loaded vinyl (MLV) or a high-density barium/composite rubber.

RISK

Trapping leaked R290 (Propane) inside a dense acoustic jacket near a crankcase heater creates a severe explosion hazard. R290 is heavier than air and will pool if the jacket is sealed.

Suggestion: Redesign the acoustic jacket to allow bottom ventilation/drainage for heavy gases and ensure all electrical components (heater) are fully ATEX/IECEx compliant.

ENGINEERING

The acoustic jacket inherently acts as a thermal blanket, directly leading to the identified thermal overload failure mode for the compressor motor.

Suggestion: Implement a thermal bridge or liquid-cooling loop at the compressor baseplate to extract heat conductively so acoustic cladding can remain intact.

ENGINEERING

Scroll bearing wear due to poor lubrication return at low inverter loads. R290 is highly miscible with POE oil, dropping its viscosity and heavily impairing oil return at low flow velocities.

Suggestion: Implement an active oil separator in the discharge line and hard-code a minimum frequency limit in the inverter firmware to maintain gas velocity.

TOLERANCES

Tuned elastomeric isolators need strict durometer tolerances to effectively dampen structural resonance.

Suggestion: Specify ±3 Shore A tolerance on the isolators and visually color-code them by weight rating on the assembly line to prevent installation mix-ups.

Strategic Recommendations

- Eliminate lead from the BOM immediately and replace with RoHS-compliant mass-loaded vinyl alternatives.
- Redesign the acoustic jacket with bottom ventilation routes to allow heavier-than-air R290 to escape safely rather than pooling near ignition sources.
- Introduce a thermally conductive base plate or heat pipe system to shunt heat out of the trapped acoustic envelope.
- Require ATEX certification for the crankcase heater and compressor terminals.
- Establish firmware safety rails enforcing a minimum compressor RPM to guarantee sufficient velocity for POE oil return.

4.2 Evaporator Coil Assembly

Lead Time: 10 weeks | Status: draft

Module Purpose

Absorbs heat from ambient air to boil low-pressure R290 liquid into vapor.

Why it matters to the system

> The coil's geometry and thermodynamic efficiency absolutely dictate the system's air-side performance, sizing, and ability to achieve the targeted MCS COP of 3.5.

Detailed Technical Description

A massive surface-area fin-and-tube heat exchanger that acts as the thermal intake for the entire system. It leverages the extremely low boiling point of the A3 R290 refrigerant to draw usable heat from ambient air, functioning effectively even at -20°C external temperatures. The dense fin pack utilizes a hydrophilic coating to promote rapid water shedding during condensation and defrost cycles.

During winter operation, ice will naturally accumulate across the coil face due to localized freezing of atmospheric humidity. This module incorporates a structurally sloped drip tray equipped with a localized trace heating cable. This ensures that when the system reverses to perform a hot-gas defrost, the melted ice flows smoothly away from the unit and does not refreeze inside the chassis, which would physically block airflow and degrade the COP.

SYSTEM INPUTS

- Ambient air
- Low-pressure R290 liquid

SYSTEM OUTPUTS

- Cooled air
- Low-pressure R290 vapor

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4.2.2 Specifications & Risk Profile

Expected Key Parts

- Hydrophilic-coated aluminum fins
- Copper tubing matrix
- Defrost drip tray
- Ice-melt heating cable

Identified Failure Modes

- Uneven frosting leading to localized airflow blockage and severe COP drop
- Galvanic corrosion over time between the copper tubes and aluminum fins
- Trace heater burnout causing absolute ice damming in the base tray

Open Questions

- Optimized fin pitch balancing UK high-humidity frost rates vs standard thermal exchange
- Structural resilience of the large unsupported coil span against commercial wind loads

4.2.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
evaporator-001	Hydrophilic-coated aluminum fins	Manufactured	-	£60.00
evaporator-002	Copper tubing matrix	Manufactured	-	£80.00
evaporator-003	Defrost drip tray	Manufactured	-	£60.00
evaporator-004	Ice-melt heating cable	Purchased	-	£40.00
		Module Total Cost:		**£240.00**

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4.2.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The evaporator coil relies on proven Cu/Al fin-and-tube architecture but requires stringent manufacturing controls due to the A3 flammable R290 refrigerant. Galvanic corrosion and trace heater chafing present distinct long-term reliability risks.

Identified Issues

RISK

R290 is an A3 highly flammable refrigerant. Any micro-leaks from the copper tube matrix brazing pose a severe safety hazard.

Suggestion: Mandate 100% helium mass spectrometry leak testing on the assembly line for all brazed joints prior to refrigerant charging.

MATERIALS

Galvanic corrosion between copper tubes and aluminum fins is highly likely in a wet, defrosting environment, even with hydrophilic coatings on the fins.

Suggestion: Apply an epoxy or polyurethane conformal coating specifically at the tube-sheet interfaces where the dissimilar metals meet.

MANUFACTURABILITY

Trace heater routing in the sloped sheet metal drip tray presents a chafing risk due to operational vibrations, potentially leading to electrical shorts or heater burnout.

Suggestion: Design stamped retention channels into the tray or use UV-rated, vibration-damping polymer clips to securely route the heating cable.

TOLERANCES

Mechanical expansion (bullet expansion) of the copper tubes to interlock with the aluminum fins requires tight tolerance control to prevent tearing the fin collars or stripping the hydrophilic coating.

Suggestion: Specify strict tooling wear limits for the mechanical expanders and validate fin collar integrity during first-article inspection.

Strategic Recommendations

- Implement automated helium trace leak detection for all copper brazings to ensure absolute R290 leak-tightness.
- Add a current-sensing circuit to the trace heating cable to detect heater burnout and alert the system before ice damming causes mechanical damage.
- Utilize pre-coated hydrophilic aluminum fin stock to streamline manufacturing, but ensure the stamping process uses optimized die clearances to minimize exposed bare aluminum edges.
- Specify heavy-duty protective sleeving or specialized retention clips for the trace heater where it contacts the sloped drip tray edge.

4.3 EC Axial Fan Module

Lead Time: 14 weeks

Status: draft

Module Purpose

Draws large volumes of ambient air across the evaporator coil efficiently and quietly.

Why it matters to the system

> Air movement dictates actual real-world heat extraction capacity, and the fans represent the primary source of tonal and broadband acoustic emissions affecting BS EN 12102 compliance.

Detailed Technical Description

Crucial to extracting 30kW of heat, this module moves massive volumes of air horizontally through the outdoor chassis. It relies on Electronically Commutated (EC) motors directly coupled to aerodynamically optimized 'owl-wing' blades. These specialized blade shapes are engineered to slice through the air while virtually eliminating trailing-edge vortex shedding, completely minimizing broadband acoustic noise to meet the <65dB threshold.

The fans are flush-mounted inside deep-drawn polymeric or sheet-metal aerodynamic shrouds. By minimizing the clearance between the blade tip and the shroud wall, backflow and internal air recirculation are eradicated. Modulated by a continuous PWM signal from the main controller, the dual fans alter their RPM precisely to match the current thermal load, ensuring they never spin faster—or louder—than absolutely necessary.

SYSTEM INPUTS

- 230V AC or 48V DC power
- PWM speed control signal

SYSTEM OUTPUTS

- Directed high-CFM airflow
- Acoustic emissions

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4.3.2 Specifications & Risk Profile

Expected Key Parts

- Biomimetic EC axial fan motors
- Aerodynamic deep-drawn shroud
- Wire bird-guards
- Anti-vibration rubber grommets

Identified Failure Modes

- Motor bearing failure leading to noisy grinding operation and eventual stall
- Fan blade mass imbalance causing severe unmitigated chassis resonance
- Ice accumulation on the blades leading to aerodynamic stall and mechanical snap

Open Questions

- Specific interaction noise (beating) between dual dynamically stacked fans
- Airflow CFM penalty caused by the required external acoustic louvers

4.3.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
fanmodule-001	Biomimetic EC axial fan motors	Purchased	-	£250.00
fanmodule-002	Aerodynamic deep-drawn shroud	Manufactured	-	£60.00
fanmodule-003	Wire bird-guards	Purchased	-	£40.00
fanmodule-004	Anti-vibration rubber grommets	Purchased	-	£40.00
		Module Total Cost:		**£390.00**

4.3.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The module utilizes highly efficient, aerodynamically proven components, but the combination of deep-drawn manufacturing methods with 'minimized' blade tip clearances poses a severe manufacturability and rubbing risk.

Identified Issues

TOLERANCES

Deep-drawn sheet metal or thermoformed polymeric shrouds cannot reliably hold the tight diametric roundness tolerances required to maintain minimal blade tip clearances without rubbing.

Suggestion: Transition to a high-stiffness injection-molded shroud (e.g., PA6-GF30) allowing precise control over the inner diameter, or increase the tip clearance and accept a marginal efficiency drop.

RISK

Ice accumulation on the blades is identified as a mechanical snap risk, but no active or passive mitigation strategy is present in the design.

Suggestion: Implement a periodic reverse-rotation de-icing cycle via the EC motor controller and specify an ice-phobic/hydrophobic coating for the blades.

ENGINEERING

Placement of the wire bird-guards too close to the inlet/outlet will create substantial turbulence, destroying the intended acoustic benefits of the biomimetic blade design.

Suggestion: Ensure the stand-off distance of the wire guards from the fan blades is optimized (typically >0.3x fan diameter) to prevent vortex interference.

MANUFACTURABILITY

Mass imbalance leading to chassis resonance is highly likely if deep-drawn shrouds flex over the anti-vibration grommets.

Suggestion: Specify factory dynamic balancing of the fan/motor assembly to ISO 1940-1 Grade G6.3 or G2.5 prior to installation.

Strategic Recommendations

- Abandon deep-drawing for the shroud; tool up for structural injection molding using glass-reinforced polymers to guarantee diametric tolerances for tight tip gaps.
- Add a reverse-spin defrost protocol to the PWM logic to shed frost and ice buildup before it solidifies into an unbalanced mass.
- Define explicit spatial constraints for the wire bird-guards to prevent turbulent noise generation.
- Enforce a mandatory dynamic balancing standard (ISO 1940-1 G2.5) for supplier fan shipments.

4.4 Hydro-Condenser Module

Lead Time: 16 weeks

Status: draft

Module Purpose

Transfers heat from the flammable high-pressure R290 into the safe indoor water circuit.

Why it matters to the system

> It determines the efficiency of thermal handover and is the central pressure vessel subject to intense European/UK safety auditing and CE/PED compliance.

Detailed Technical Description

This sub-assembly acts as the critical thermal bulkhead between the hazardous external R290 refrigerant loop and safely heated water entering the commercial space. At its core is a commercial-grade 316 stainless steel Brazed Plate Heat Exchanger (BPHE). This PED-compliant BPHE utilizes an asymmetrical channel design to minimize the volume of A3 refrigerant required inside the plate matrix while simultaneously maximizing the heat transfer coefficient to the water side.

Included in this module are the primary wet components that condition the building's water loop. A highly efficient inverter-driven circulation pump modulates water flow, aligning the GPM (gallons per minute) perfectly with the instantaneous thermal output of the compressor. Critical thermal safety hardware, such as flow switches, prevent catastrophic high-pressure faults by refusing to permit compressor operation if water flow is obstructed.

SYSTEM INPUTS

- High-pressure R290 vapor
- Cool return commercial utility water

SYSTEM OUTPUTS

- High-pressure R290 liquid
- Hot supply water (up to 75°C)

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4.4.2 Specifications & Risk Profile

Expected Key Parts

- Stainless steel brazed plate heat exchanger
- Inverter water circulation pump
- Water flow switch
- Pressure relief valve

Identified Failure Modes

- Internal plate rupture cross-contaminating R290 straight into the commercial building water loop
- Calcareous scale build-up inside the water channels severely dropping heat transfer
- Pump cavitation operating at high flow-rates and maximum 75°C flow temperatures

Open Questions

- Exactly mapped pressure drop across asymmetrical BPHE plates with heavy glycol mixtures
- Vibrational impact of the high-speed water pump on adjacent copper refrigerant lines

4.4.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
hydrocondenser-001	Stainless steel brazed plate heat exchanger	Purchased	-	£600.00
hydrocondenser-002	Inverter water circulation pump	Purchased	-	£250.00
hydrocondenser-003	Water flow switch	Purchased	-	£40.00
hydrocondenser-004	Pressure relief valve	Purchased	-	£80.00
			Module Total Cost:	**£970.00**

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4.4.4 Engineering Review

Review by Fang

FAIL

Verdict Summary

The current design poses a catastrophic safety risk due to the potential of highly flammable A3 (R290) refrigerant cross-contaminating the indoor commercial water loop. A single-wall BPHE is insufficient for this application; a double-wall vented system must be implemented.

Identified Issues

RISK

Internal plate rupture can inject high-pressure R290 (Propane) directly into the indoor water circuit, creating a severe explosion and fire hazard inside the commercial building.

Suggestion: Replace the single-wall BPHE with a double-wall Brazed Plate Heat Exchanger equipped with an atmospheric vent and electronic refrigerant leak detector.

ENGINEERING

Pump cavitation operating at 75°C flow temperatures. At this temperature, the vapor pressure of water is high, significantly reducing the Net Positive Suction Head Available (NPSHa).

Suggestion: Ensure the circulation pump is placed on the cooler return water side of the heat exchanger and establish minimum system static pressure requirements.

MATERIALS

Calcareous scale build-up inside the narrow asymmetrical channels of the BPHE at 75°C will severely degrade heat transfer and increase pressure drop.

Suggestion: Integrate isolation/service valves and dedicated flush ports into the module manifold to allow for routine acid descaling maintenance.

MANUFACTURABILITY

Assembly of the R290 and water lines to the BPHE requires strict process control to avoid warping the heat exchanger or causing micro-leaks at brazed joints.

Suggestion: Implement helium leak testing on the R290 side and hydrostatic pressure testing on the water side prior to final module sign-off.

Strategic Recommendations

- Immediately halt the single-wall BPHE design and specify a double-wall BPHE to provide a mechanical barrier and safe venting path against R290 cross-contamination.
- Add a dedicated A3 gas leak detection sensor within the module housing, particularly near the double-wall vent.
- Reconfigure the plumbing layout to locate the inverter pump on the cooler return water line to improve NPSHa and eliminate 75°C cavitation risks.
- Add standard flush ports (with isolation valves) to the water side of the module for mandatory field de-scaling operations.

4.5 Refrigerant Flow Control Circuit

Lead Time: 8 weeks

Status: draft

Module Purpose

Directs, expands, and conditions the flow of R290 safely throughout the unit.

Why it matters to the system

> The dynamic precision of the EEV determines the heat pump's operational efficiency envelope and actively prevents devastating compressor liquid-slugging.

Detailed Technical Description

Functionally acting as the vascular system, this module manages the state, pressure, and direction of the R290 fluid. The Electronic Expansion Valve (EEV) dynamically restricts flow to precisely drop the pressure of the R290 liquid, causing it to flash into a 2-phase mixture before entering the evaporator. Micro-stepping control of the EEV maintains perfect system superheat, protecting the compressor from liquid destruction while maximizing energy extraction.

The large 4-way reversing valve physically switches the entire thermodynamic cycle. During deep winter, it channels hot discharge gas straight into the outdoor evaporator coil to melt accumulated ice. Embedded throughout this circuit are intrinsically safe high and low-pressure transducers, constantly reporting pressures back to the ATEX board to operate safely within the flammable A3 envelope.

SYSTEM INPUTS

- High-pressure R290 liquid
- Control board signals

SYSTEM OUTPUTS

- Low-pressure R290 liquid/gas mix
- Reversed flow path (Defrost)

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4.5.2 Specifications & Risk Profile

Expected Key Parts

- Electronic Expansion Valve (EEV)
- 4-way reversing valve
- High/Low pressure transducers
- Filter drier core
- Sight glass

Identified Failure Modes

- 4-way reversing valve slider jamming mid-stroke causing immense internal capability loss
- Stepper motor detachment from EEV body leading to unregulated flood-back
- Filter drier core disintegrating and sending desiccant beads into the EEV orifice

Open Questions

- Real-world phase shift response time required during rapid defrost valve transitions
- Amount of vibration-induced fatigue settling on the complex nest of copper u-bends

4.5.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
refrigerantcircuit-001	Electronic Expansion Valve (EEV)	Purchased	-	£150.00
refrigerantcircuit-002	4-way reversing valve	Purchased	-	£80.00
refrigerantcircuit-003	High/Low pressure transducers	Purchased	-	£60.00
refrigerantcircuit-004	Filter drier core	Purchased	-	£40.00
refrigerantcircuit-005	Sight glass	Purchased	-	£40.00
			Module Total Cost:	**£370.00**

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4.5.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The baseline thermodynamic design is sound, but employing highly flammable R290 (A3) requires elimination of unnecessary leak points like sight glasses and demands aggressive contamination control during manufacturing.

Identified Issues

RISK

Inclusion of a sight glass in a sealed R290 (Propane) system adds an unacceptable and unnecessary glass-to-metal seal leak failure point.

Suggestion: Remove the sight glass for production OEM units. If diagnostics are needed, rely on the intrinsic high/low pressure transducers and EEV telemetry.

MANUFACTURABILITY

4-way reversing valves are highly susceptible to jamming due to cupric oxide flaking (scale) introduced during copper tube brazing.

Suggestion: Mandate a continuous nitrogen purge during all brazing operations and implement strict workplace cleanliness requirements.

MATERIALS

Filter drier core disintegration threatens the EEV orifice. Loose-fill desiccant beads can crush under vibration.

Suggestion: Specify a 100% solid-core (molded) filter drier designed for high-vibration environments, compatible with R290 and the selected POE/PAG oil.

ENGINEERING

EEV stepper motor detachment under operational vibration leads to severe system flooding and compressor damage.

Suggestion: Design a positive locking mechanism (e.g., locking tabs, safety wire, or vibration-resistant retaining clips) to secure the EEV stator to the valve body.

Strategic Recommendations

- Eliminate the sight glass from the bill of materials to reduce A3 flammability risk.
- Implement an end-of-line mass spectrometer Helium leak detection process for the entire brazed circuit.
- Switch all filter driers to high-density solid-core variants rather than spun-bead models.
- Revise Assembly SOPs to require strict flowing nitrogen (N2) inside piping during any brazing to prevent oxidation layer formation.
- Introduce a vibration test (shake table) for early production samples to validate EEV stepper motor clip retention.

4.6 3-Phase Power Inverter Drive

Lead Time: 20 weeks

Status: draft

Module Purpose

Converts strict grid AC power to variable-frequency AC to modulate compressor speeds safely.

Why it matters to the system

> Modulation via the inverter is absolutely necessary for part-load efficiency (achieving high COP) and G99 grid compliance regarding harmonic distortion and starting currents.

Detailed Technical Description

Responsible for managing the 30kW unit's intense electrical demands, this sub-assembly draws 400V 3-phase commercial grid power, rectifies it to a flat high-voltage DC bus, and then utilizes Insulated Gate Bipolar Transistors (IGBTs) to invert the power back into a synthesized, variable-frequency AC waveform. This technology abolishes hard-starts, reducing electrical grid transient spikes, and allows the compressor to smoothly ramp its output to match commercial heating demands seamlessly.

Due to the extreme spatial constraints (1200x800x1400mm enclosure), air-cooling massive heat sinks for the IGBTs is spatially prohibited and acoustically detrimental. Therefore, the power electronics are mated to an integrated liquid-cooled cold plate, transferring their substantial waste heat directly into the primary water return line. This recovers energy back into the heating circuit while maintaining the electronics well below their thermal derating curves.

SYSTEM INPUTS

- 400V 3-Phase AC 50Hz

SYSTEM OUTPUTS

- Variable frequency 3-Phase AC
- Regulated DC Bus voltage
- Harmonic mitigated power

This section was produced by google/gemini-3.1-pro-preview via OpenRouter. Source: LLM research synthesis.

4.6.2 Specifications & Risk Profile

Expected Key Parts

- IGBT switching bridge module
- High-capacity DC link capacitors
- Active EMI/RFI filter block
- Liquid-cooled aluminum cold plate

Identified Failure Modes

- IGBT shoot-through and catastrophic shorting due to compromised cold-plate thermal paste
- DC link capacitor internal venting triggered by severe commercial grid voltage transients
- EMI filter saturation leading to radiated emissions disrupting BMS and nearby commercial equipment

Open Questions

- Exact required thermal dissipation profile from the cold plate into low-load return water
- Total Harmonic Distortion (THD) footprints at very low rotational speeds

4.6.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
inverterdrive-001	IGBT switching bridge module	Purchased	-	£40.00
inverterdrive-002	High-capacity DC link capacitors	Purchased	-	£40.00
inverterdrive-003	Active EMI/RFI filter block	Purchased	-	£40.00
inverterdrive-004	Liquid-cooled aluminum cold plate	Manufactured	-	£80.00
			Module Total Cost:	**£200.00**

This section was produced by google/gemini-3.1-pro-preview via OpenRouter. Source: LLM research synthesis.

4.6.4 Engineering Review

Review by Fang

WARN

Verdict Summary

Using the primary heating return loop to liquid-cool the IGBTs is an excellent space-saving and energy-recovery concept, but exposes the 400V electronics to critical thermal runaways and catastrophic galvanic corrosion if the water loop is not strictly isolated and controlled.

Identified Issues

ENGINEERING

Using the primary water return line for cooling assumes the return water is always cold enough to act as a heat sink. If the commercial heating loop runs hot (e.g., 60°C+ return), the IGBTs will lose their thermal delta and fail catastrophically under load.

Suggestion: Conduct a strict thermodynamic profile of the max return water temperature. Implement software interlocks to derate compressor speed if cold-plate thermistors exceed safe limits.

MATERIALS

Circulating commercial heating system water directly through an aluminum cold plate will rapidly induce galvanic corrosion, scaling, and eventual coolant leaks into the 400V/30kW drive.

Suggestion: Use a stainless-steel tube embedded in an aluminum plate (tube-in-plate) or utilize a CuNi sub-loop with an intermediate plate heat exchanger.

RISK

DC link capacitors venting due to grid transients implies inadequate frontend overvoltage protection. Venting inside a sealed enclosure can lead to explosive failure and total unit loss.

Suggestion: Implement heavily-rated Metal Oxide Varistors (MOVs) and transient voltage suppression (TVS) arrays on the AC input upstream of the rectifier/EMI filter.

MANUFACTURABILITY

Relying on 'thermal paste' for the large surface area of high-power IGBT modules is a massive yield and reliability vulnerability. Manual application causes voids, and thermal cycling causes 'pump-out'.

Suggestion: Switch to pre-cut Phase Change Material (PCM) pads or extreme-durability thermal gap pads that guarantee uniform interface thickness.

TOLERANCES

Liquid-cooled cold plates require extreme flatness to mate effectively with bare IGBT substrates. Bowing under mounting torque will create thermal air gaps.

Suggestion: Specify a CNC fly-cut finish with a flatness tolerance of <0.05mm/100mm and outline a specific star-pattern torque sequence for assembly.

Strategic Recommendations

- Redesign the cold plate to use corrosion-resistant fluid paths (e.g., embedded stainless steel tubing) rather than raw aluminum wetted surfaces.
- Replace wet thermal paste processes with automated Phase Change Material (PCM) phase pads to eliminate assembly voids and long-term pump-out.
- Add dedicated surge arrestors (MOVs) upstream of the drive to clamp commercial grid spikes before they stress the DC link capacitors.
- Integrate high-speed thermistors directly into the IGBT baseplates, with hard-wired interlocks to shut down the PWM signal if the primary return water runs too hot.

4.7 ATEX Central Control Unit

Lead Time: 12 weeks

Status: draft

Module Purpose

Acts as the logical brain executing thermodynamic algorithms, BMS communications, and absolute safety isolation.

Why it matters to the system

> It houses the core IP (efficiency algorithms) and guarantees safe failure modes according to BS EN 378 codes for dangerous flammable refrigerants.

Detailed Technical Description

The logic center of the air source heat pump hosts the proprietary thermodynamic algorithms that continuously balance fan speeds, EEV positioning, and compressor hertz to meet MCS certifications. Crucially, as the system functions around highly flammable A3 propane, this entire module must be aggressively hardened against ignition hazard. The sub-assembly is sealed inside an ATEX-compliant, intrinsically safe enclosure that prevents any arcing relay from igniting atmospheric leaks.

It features constant interoperability with building management systems via Modbus/BACnet for commercial operators. Furthermore, it incorporates an active LEL (Lower Explosive Limit) R290 sniffer inside the chassis. If the sniffer detects even a slight concentration of escaped propane, this module defaults to an immediate hardware interrupt, shutting off potential spark sources while forcing the axial fans to 100% capacity to purge the explosive gas safely.

SYSTEM INPUTS

- Thermistor and pressure arrays
- BMS/Modbus telemetry commands
- R290 combustible gas sniffer

SYSTEM OUTPUTS

- PWM to system fans
- EEV step pulses
- Target inverter frequencies
- Safety contactor cut-offs

This section was produced by google/gemini-3.1-pro-preview via OpenRouter. Source: LLM research synthesis.

4.7.2 Specifications & Risk Profile

Expected Key Parts

- Central processor PCB
- ATEX-rated IP65 enclosure
- Spark-free isolation relays
- Internal R290 leak detector

Identified Failure Modes

- Relay arcing bypassing the ATEX hermetic seal, providing an ignition source
- Analog-to-Digital Converter drift causing poor evaluation of superheat thermodynamics
- Firmware lockup leaving the system stranded in an active defrost cycle

Open Questions

- Lifespan and false-positive rate of the integrated R290 sensor in humid environments
- Total execution latency during rapid multi-variable PID control loops

This section was produced by google/gemini-3.1-pro-preview via OpenRouter. Source: LLM research synthesis.

4.7.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
controlbox-001	Central processor PCB	Purchased	-	£40.00
controlbox-002	ATEX-rated IP65 enclosure	Purchased	-	£120.00
controlbox-003	Spark-free isolation relays	Purchased	-	£40.00
controlbox-004	Internal R290 leak detector	Purchased	-	£40.00
			Module Total Cost:	**£240.00**

This section was produced by google/gemini-3.1-pro-preview via OpenRouter. Source: LLM research synthesis.

4.7.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The control unit demonstrates a robust safety philosophy for R290 integration, but features critical contradictions regarding ATEX protection methods and thermal management that must be resolved before production.

Identified Issues

ENGINEERING

Contradictory ATEX protection concepts. 'Intrinsically safe' (Ex i) limits circuit energy and cannot be applied to an enclosure housing power-switching relays. The enclosure must be rated Flameproof (Ex d), Encapsulated (Ex m), or Non-Arcing (Ex nC).

Suggestion: Reclassify the enclosure certification pathway to Ex d (Flameproof) or use Ex m (Encapsulated) solid-state relays.

RISK

Emergency purge logic relies on powering axial fans to 100% when an LEL event is detected. If the fans or their contactors are not ATEX-certified, this purge action introduces an ignition source directly into the gas cloud.

Suggestion: Mandate that the axial fans and their isolated drive circuits carry ATEX Zone 2 (or better) certification and use brushless, spark-free motors.

MANUFACTURABILITY

ATEX enclosure sealing requires specialized Ex-rated cable glands for all I/O (sensors, Modbus, power). Improper torque or sealing of these glands during assembly compromises the Ex rating.

Suggestion: Develop specialized assembly SOPs outlining exact torque specifications, thread engagement rules, and QA potting/sealing verification for all cable entries.

ENGINEERING

Trapped heat inside a hermetically sealed IP65/ATEX enclosure will accelerate ADC drift and degrade PCB longevity, especially if using solid-state isolation relays.

Suggestion: Integrate a passive thermal bridge to the enclosure chassis, specify high-temperature rated components, and use a precision voltage reference for the A/D circuits.

ENGINEERING

Firmware lockup leaving the system in an active defrost cycle poses a severe physical equipment risk (evaporator freeze block or compressor floodback).

Suggestion: Implement a discrete, external hardware Watchdog Timer (WDT) that physically interrupts power to actuators if the MCU halts.

Strategic Recommendations

- Redefine the ATEX protection concept from 'Intrinsically Safe' to 'Flameproof' (Ex d) or 'Restricted Breathing' (Ex nR) suitable for power relay applications.
- Verify that the axial purge fans and their respective power lines are completely spark-free and Ex-certified for duty in a flammable atmosphere.
- Implement an external hardware watchdog timer to safely unlatch all structural relays in the event of firmware lockup.
- Design a thermal pathway (e.g., thermal potting or internal heat sinking to the metal Ex enclosure) to prevent internal heat buildup and subsequent ADC drift.
- Introduce a strict Quality Assurance gating step on the assembly line specifically for torquing and verifying Ex-rated cable glands.

4.8 Acoustic Structural Casing

Lead Time: 6 weeks

Status: draft

Module Purpose

Provides environmental protection, structural lifting integrity, and comprehensive acoustic mitigation.

Why it matters to the system

> Limits the physical footprint, ensures the unit doesn't act as a resonant drum, and provides the fundamental ATEX ventilation requirements to pass EN 378 safety standards.

Detailed Technical Description

The heavy-duty exoskeleton rigorously adhering to the 1200mm x 800mm x 1400mm dimensions. It is built upon a substantially stiffened base pan that absolutely guarantees structural coherence during installation via crane or forklift. To achieve the phenomenally quiet sub-65dB specification, the internal panels are densely clad with acoustic metamaterials—a sandwich of closed-cell dampening foam and mass-loaded exterior vinyl to absorb both low-frequency structural hums and high-frequency hisses.

Acoustics and flammable refrigerant physics are carefully balanced in this design. R290 gas is heavier than air; therefore, the lowest point of the chassis consists of grated, unrestricted drop-out voids that permit leaked propane to disperse into the wind safely. Exhaust louvers are angled with baffles that strip energy out of exiting sound waves while remaining completely weatherproof against the damp UK climate.

SYSTEM INPUTS

- Environmental impacts (wind, rain, UV)
- Internal system acoustic waves

SYSTEM OUTPUTS

- Muffled exhaust noise
- Safe natural drop-out paths for leaked R290

4.8.2 Specifications & Risk Profile

Expected Key Parts

- Heavy-gauge folded steel base pan
- Acoustic composite-lined side panels
- Weatherproof exhaust louvers
- Forklift channels and lifting eyes

Identified Failure Modes

- Insufficient base pan stiffness leading to copper pipe fracture during lifting
- Acoustic foam saturation from rain ingress leading to severe rotting and weight gain
- Corrosion fatigue at key panel joints due to commercial shoreline applicability

Open Questions

- Exact wet-weight center of gravity impacting lifting procedures
- Acoustic flanking pathways escaping through the chassis edge seams

4.8.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
chassis-001	Heavy-gauge folded steel base pan	Manufactured	-	£60.00
chassis-002	Acoustic composite-lined side panels	Manufactured	-	£60.00
chassis-003	Weatherproof exhaust louvers	Manufactured	-	£60.00
chassis-004	Forklift channels and lifting eyes	Manufactured	-	£60.00
			Module Total Cost:	**£240.00**

This section was produced by google/gemini-3.1-pro-preview via OpenRouter. Source: LLM research synthesis.

4.8.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The casing's exact standard logistical footprint and R290 safety integration are well conceptualized, but the reliance on standard heavy-gauge steel poses a critical corrosion risk in the target shoreline environments. Additionally, bottom grating will create acoustic flanking paths that threaten the sub-65dB requirement unless baffled.

Identified Issues

MATERIALS

Commercial shoreline applicability exposes a standard folded steel base pan to rapid salt-spray corrosion, leading to joint fatigue and structural failure during handling.

Suggestion: Specify marine-grade materials (e.g., Aluzinc, Magnelis, or 316L Stainless Steel) combined with a C5-M rated powder coating for all exterior and structural components.

ENGINEERING

Unrestricted drop-out grated voids at the base for R290 dispersion will allow low-frequency noise to escape downwards and reflect off the installation pad, compromising the sub-65dB spec.

Suggestion: Design an acoustically treated, slanted labyrinth or a baffled channel for the bottom void that allows heavier-than-air gas to cascade out while absorbing downward sound waves.

MANUFACTURABILITY

Acoustic metamaterial sandwiches (closed-cell foam + MLV) are prone to delamination and moisture wicking at the exposed cut edges, despite the closed-cell nature of the main foam body.

Suggestion: Implement an edge-sealing manufacturing step (e.g., heat sealing, foil tape wrapping, or RTV silicone edges) for all acoustic panel cutouts to prevent rotting and weight gain.

ENGINEERING

Lifting eyes directly attached to folded sheet metal without dedicated structural nodes will cause localized yielding and warp the base pan when hoisted by a crane.

Suggestion: Weld 5mm+ minimum thickness load-spreading gusset plates at the four lifting eye nodes and tie them directly into the heavy-gauge forklift channels.

RISK

Large grated voids on the lowest point will invite rodents or insects, potentially leading to nesting inside the acoustic foam.

Suggestion: Ensure the grate aperture is no larger than 5mm, or add a secondary stainless-steel physical pest mesh across all drop-out voids.

Strategic Recommendations

- Upgrade the material specification of the base pan and exterior panels to Magnelis or 316L Stainless steel to mitigate shoreline corrosion fatigue.
- Redesign the R290 drop-out bottom voids to incorporate acoustic baffles that prevent noise from bouncing off concrete mounting pads while still allowing propane egress.
- Mandate edge-sealing on the BOM for all cut-to-size acoustic dampening foam/MLV composite panels to prevent moisture ingress and saturation.
- Perform structural FEA strictly on the crane-lifting scenario to size welded gussets at the lifting eye nodes, preventing base pan copper fracture.
- Integrate a sub-5mm pest screen above the R290 dispersion grates.

5. BILL OF MATERIALS

5.1 BOM — R290 Scroll Compressor Assembly

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
compressor-001	Inverter scroll compressor	purchased_cots / varies	Purchased	-	£800.00
compressor-002	Acoustic compressor jacket	purchased_cots / varies	Purchased	-	£800.00
compressor-003	Anti-vibration elastomeric mounts	purchased_cots / varies	Purchased	-	£40.00
compressor-004	Crankcase heater	purchased_cots / varies	Purchased	-	£40.00

5.2 BOM — Evaporator Coil Assembly

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
evaporator-001	Hydrophilic-coated aluminum fins	sheet_metal / varies	Manufactured	-	£60.00
evaporator-002	Copper tubing matrix	cnc / varies	Manufactured	-	£80.00
evaporator-003	Defrost drip tray	sheet_metal / varies	Manufactured	-	£60.00
evaporator-004	Ice-melt heating cable	purchased_cots / varies	Purchased	-	£40.00

5.3 BOM — EC Axial Fan Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
fanmodule-001	Biomimetic EC axial fan motors	purchased_cots / varies	Purchased	-	£250.00
fanmodule-002	Aerodynamic deep-drawn shroud	sheet_metal / varies	Manufactured	-	£60.00
fanmodule-003	Wire bird-guards	purchased_cots / varies	Purchased	-	£40.00
fanmodule-004	Anti-vibration rubber grommets	purchased_cots / varies	Purchased	-	£40.00

5.4 BOM — Hydro-Condenser Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
hydrocon-denser-001	Stainless steel brazed plate heat exchanger	purchased_cots / varies	Purchased	-	£600.00
hydrocon-denser-002	Inverter water circulation pump	purchased_cots / varies	Purchased	-	£250.00
hydrocon-denser-003	Water flow switch	purchased_cots / varies	Purchased	-	£40.00
hydrocon-denser-004	Pressure relief valve	purchased_cots / varies	Purchased	-	£80.00

5.5 BOM — Refrigerant Flow Control Circuit

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
refrigerantcircuit-001	Electronic Expansion Valve (EEV)	purchased_cots / varies	Purchased	-	£150.00
refrigerantcircuit-002	4-way reversing valve	purchased_cots / varies	Purchased	-	£80.00
refrigerantcircuit-003	High/Low pressure transducers	purchased_cots / varies	Purchased	-	£60.00
refrigerantcircuit-004	Filter drier core	purchased_cots / varies	Purchased	-	£40.00
refrigerantcircuit-005	Sight glass	purchased_cots / varies	Purchased	-	£40.00

5.6 BOM — 3-Phase Power Inverter Drive

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
inverterdrive-001	IGBT switching bridge module	purchased_cots / varies	Purchased	-	£40.00
inverterdrive-002	High-capacity DC link capacitors	purchased_cots / varies	Purchased	-	£40.00
inverterdrive-003	Active EMI/RFI filter block	purchased_cots / varies	Purchased	-	£40.00
inverterdrive-004	Liquid-cooled aluminum cold plate	cnc / varies	Manufactured	-	£80.00

5.7 BOM — ATEX Central Control Unit

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
controlbox-001	Central processor PCB	purchased_cots / varies	Purchased	-	£40.00
controlbox-002	ATEX-rated IP65 enclosure	purchased_cots / varies	Purchased	-	£120.00
controlbox-003	Spark-free isolation relays	purchased_cots / varies	Purchased	-	£40.00
controlbox-004	Internal R290 leak detector	purchased_cots / varies	Purchased	-	£40.00

5.8 BOM — Acoustic Structural Casing

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
chassis-001	Heavy-gauge folded steel base pan	sheet_metal / varies	Manufactured	-	£60.00
chassis-002	Acoustic composite-lined side panels	sheet_metal / varies	Manufactured	-	£60.00
chassis-003	Weatherproof exhaust louvers	sheet_metal / varies	Manufactured	-	£60.00
chassis-004	Forklift channels and lifting eyes	sheet_metal / varies	Manufactured	-	£60.00

6. Cost Waterfall & Economics

UNIT COST	TARGET CEILING	HEADROOM
£6,495.00	None	N/A

Make vs Buy Split

PURCHASED (BUY) TOTAL	CUSTOM (MAKE) TOTAL	MAKE %
£3,750.00	£580.00	8.9%

Per-Module Roll-up

MODULE	% OF UNIT	COST
Module compressor	38.8%	£2,520.00
Module evaporator	5.5%	£360.00
Module fanmodule	9.0%	£585.00
Module hydrocondenser	22.4%	£1,455.00
Module refrigerantcircuit	8.5%	£555.00
Module inverterdrive	4.6%	£300.00
Module controlbox	5.5%	£360.00
Module chassis	5.5%	£360.00

Non-recurring engineering and overheads

Total non-recurring engineering (Tooling, Certs)	**£4,800.00**
Complexity Overhead Multiplier	**1.5×**

Grand Total (Unit plus Non-recurring Engineering) **£11,295.00**

Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

7.1 R290 Scroll Compressor Assembly Risks

Risk 1: Liquid slugging on cold start

Base Score: S:5 × L:2 = 10

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Mandate crankcase heater operation 12 hours prior to start

Owner: Thermal Engineer

Risk 2: Vibration fatigue of copper output lines

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Incorporate vibration elimination loops in discharge pipework

Owner: Mechanical Engineer

7.1 R290 Scroll Compressor Assembly Risks

Risk 3: Compressor motor overheating

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Discharge temperature thermistor active limiting

Owner: Controls Engineer

7.2 Evaporator Coil Assembly Risks

Risk 4: Ice damming in drip tray

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Increase trace heater wattage and optimize pan slope

Owner: Mechanical Engineer

Risk 5: Micro-leak at hairpin braze joints

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Automated helium leak testing on 100% of sub-assemblies

Owner: Manufacturing Engineer

7.2 Evaporator Coil Assembly Risks

Risk 6: Coil fin damage during transport

Base Score: S:3 × L:4 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Add rigid protective grilles during assembly and transport

Owner: Packaging Engineer

7.3 EC Axial Fan Module Risks

Risk 7: Acoustic resonance matching chassis frequency

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Implement software RPM skip-bands to avoid resonant frequencies

Owner: Acoustics Engineer

Risk 8: Blade fracture from ice impact

Base Score: S:4 × L:2 = 8

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Material selection of toughened glass-reinforced polymer

Owner: Mechanical Engineer

7.3 EC Axial Fan Module Risks

Risk 9: Motor failure due to water ingress

Base Score: S:3 × L:2 = 6

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Specify IP55 or higher rated EC motors

Owner: Electrical Engineer

7.4 Hydro-Condenser Module Risks

Risk 10: BPHE internal rupture

Base Score: S:5 × L:1 = 5

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Mandate strictly oversized PED safety margins and twin-wall plate options

Owner: Compliance Engineer

Risk 11: Water-side freezing when powered off

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Logic implementation to pulse water pump periodically at ambients below 2°C

Owner: Controls Engineer

7.4 Hydro-Condenser Module Risks

Risk 12: Flow switch sticking closed

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Implement secondary software checking of entering/leaving water temp deltas

Owner: Controls Engineer

7.5 Refrigerant Flow Control Circuit Risks

Risk 13: Refrigerant micro-leak at braze joints

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Minimize joints and utilize automated induction brazing

Owner: Manufacturing Engineer

Risk 14: Valve sticking during defrost

Base Score: S:3 × L:2 = 6

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Ensure correct valve sizing and pressure differential across the pilot

Owner: Thermal Engineer

7.5 Refrigerant Flow Control Circuit Risks

Risk 15: Pressure transducer drift

Base Score: S:3 × L:3 = 9

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Specify high-accuracy automotive-grade hermetic sensors

Owner: Electrical Engineer

7.6 3-Phase Power Inverter Drive Risks

Risk 16: Thermal overload of IGBTs

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Install redundant NTC thermistors strictly glued to the power module

Owner: Electrical Engineer

Risk 17: Grid non-compliance due to EMI

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Extensive pre-compliance testing with multi-stage choke designs

Owner: Compliance Engineer

7.6 3-Phase Power Inverter Drive Risks

Risk 18: Condensation on cold plate shorting traces

Base Score: S:5 × L:3 = 15

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Full conformal coating of PCB and rigid physical potting

Owner: Electrical Engineer

7.7 ATEX Central Control Unit Risks

Risk 19: Ignition of pooled R290

Base Score: S:5 × L:1 = 5

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Strict ATEX Zone 2 design rules for all internal control electronics

Owner: Compliance Engineer

Risk 20: False positive leak detection causing lockout

Base Score: S:3 × L:4 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Dual discrete R290 sensors requiring cross-validation before lock

Owner: Controls Engineer

7.7 ATEX Central Control Unit Risks

Risk 21: BMS communication loss

Base Score: S:2 × L:3 = 6

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Watchdog timers for Modbus stack with safe autonomous fallback mode

Owner: Software Engineer

7.8 Acoustic Structural Casing Risks

Risk 22: Frame warping during forklift handling

Base Score: S:4 × L:2 = 8

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Incorporate cross-braced steel channels parallel to forks

Owner: Mechanical Engineer

Risk 23: Acoustic leakage over 65dB limit

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Extensive hemi-anechoic chamber testing on pre-production chassis

Owner: Acoustics Engineer

7.8 Acoustic Structural Casing Risks

Risk 24: R290 pooling in base pan recesses

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Fluid simulation to guarantee 100% downward clearance of heavier-than-air gases

Owner: Compliance Engineer

8.1 Inverter scroll compressor (compressor-001)

8.2 Acoustic compressor jacket (compressor-002)

8.3 Anti-vibration elastomeric mounts (compressor-003)

8.4 Crankcase heater (compressor-004)

8.5 Hydrophilic-coated aluminum fins (evaporator-001)

8.6 Copper tubing matrix (evaporator-002)

8.7 Defrost drip tray (evaporator-003)

8.8 Ice-melt heating cable (evaporator-004)

8.9 Biomimetic EC axial fan motors (fanmodule-001)

8.10 Aerodynamic deep-drawn shroud (fanmodule-002)

8.11 Wire bird-guards (fanmodule-003)

8.12 Anti-vibration rubber grommets (fanmodule-004)

8.13 Stainless steel brazed plate heat exchanger (hydrocondenser-001)

8.14 Inverter water circulation pump (hydrocondenser-002)

8.15 Water flow switch (hydrocondenser-003)

8.16 Pressure relief valve (hydrocondenser-004)

8.17 Electronic Expansion Valve (EEV) (refrigerantcircuit-001)

8.18 4-way reversing valve (refrigerantcircuit-002)

8.19 High/Low pressure transducers (refrigerantcircuit-003)

8.20 Filter drier core (refrigerantcircuit-004)

8.21 Sight glass (refrigerantcircuit-005)

8.22 IGBT switching bridge module (inverterdrive-001)

8.23 High-capacity DC link capacitors (inverterdrive-002)

8.24 Active EMI/RFI filter block (inverterdrive-003)

8.25 Liquid-cooled aluminum cold plate (inverterdrive-004)

8.26 Central processor PCB (controlbox-001)

8.27 ATEX-rated IP65 enclosure (controlbox-002)

8.28 Spark-free isolation relays (controlbox-003)

8.29 Internal R290 leak detector (controlbox-004)

8.30 Heavy-gauge folded steel base pan (chassis-001)

8.31 Acoustic composite-lined side panels (chassis-002)

8.32 Weatherproof exhaust louvers (chassis-003)

8.33 Forklift channels and lifting eyes (chassis-004)

9. Audit Log

> Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	**local-1777895137923**
Research Result	**Generated**
System Modules	**8 Modules**
Bill of Materials Lines	**33 Parts**
Cost Analysis	**Complete**
Specialist Reviews	**8 Completed**
Supplier Shortlists	**33 Matched**

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction
Modules	LLM	Unknown LLM	Gemini 3.1 Pro — module decomposition
BOM	DETERMINISTIC	deterministic	Deterministic expansion from Max keyParts
BOM	LLM	Unknown LLM	Gemini 3.1 Pro — gap-fill for missing standard hardware
Cost	DETERMINISTIC	deterministic	Domain overhead multiplier model
Suppliers	SEARCH	search	Brave Search API — commercial supplier matching
Reviews	LLM	Unknown LLM	Gemini 3.1 Pro — Fang engineering review
Proofreader	LLM	Unknown LLM	Gemini 3.1 Pro — cross-module consistency check

11. Quality Scores

Automated self-critique mechanism evaluating the completeness, realism, and engineering validity of each section.

Brief

Score: 6/10

EVALUATION REASONS

- Strong strategic foundation with a clear mission, UK-specific target market, and compelling regulatory timing rationale.
- Complete failure to provide quantitative engineering constraints and technical parameters, leaving multiple essential fields blank or 'not set'.
- The brief demonstrates strong clarity in mission and customer segmentation, with a solid market timing rationale, but is severely lacking in engineering constraints and technical specifics, reducing its overall completeness and utility for development.
- While key business aspects are well-articulated, the absence of numerical constraints (e.g., cost, mass) and technical details (e.g., processes, materials) indicates gaps that could hinder practical implementation.
- The mission and customer identification are strong and well-targeted, providing a clear focus for the product.

IMPROVEMENT SUGGESTIONS

- Add a validation layer in the generation pipeline that rejects or re-prompts if constraint fields (Cost, Mass, Quantity) evaluate to 'not set', null, or empty.
- Update the extraction LLM prompt to explicitly mandate the generation of realistic estimates for 'Material', 'Process', 'Tolerance', and 'Compliance' if they are missing from the raw input.
- Implement validation rules in the pipeline to require or prompt for the setting of cost ceiling, mass, dimensions, and quantity fields before the brief is finalized, ensuring all engineering constraints are addressed.
- Add a mandatory checklist or template in the pipeline that includes fields for materials, processes, tolerance, and operating parameters, with guidance to fill them in to enhance technical specificity.
- Introduce a mandatory 'Constraints' block with minimum requirements for cost ceiling, max mass, dimensions, and target quantity; flag missing values as errors.

Regulatory

Score: 7/10

EVALUATION REASONS

- Substantial content
- Contains numerical data
- Contains units of measurement

Sizing

Score: 6/10

EVALUATION REASONS

- Contains numerical data
- Contains units of measurement

Modules

Score: 8/10

EVALUATION REASONS

- Substantial content
- Comprehensive content
- Contains numerical data
- Contains units of measurement

BOM

Score: 5/10

EVALUATION REASONS

- The BOM successfully achieves a realistic modular breakdown with appropriate commercial off-the-shelf identification.
- The data quality is heavily degraded by explicit placeholder values, including 0kg masses and repeating cookie-cutter costs (40, 60).
- Engineering specificity is low due to the lack of actual material grades and occasional incorrect manufacturing process assignments.
- Complete part coverage ensures all modules are addressed.
- Lack of material specifications reduces technical accuracy and traceability.

IMPROVEMENT SUGGESTIONS

- Fix the mass calculation bug in the pipeline that is causing all parts to output as 0kg.
- Update the generation prompt to enforce specific engineering material grades (e.g., 'Stainless Steel 304' instead of just 'Stainless steel').
- Replace static cost generation with a parametric cost estimation function based on part mass, material grade, and manufacturing process to eliminate repetitive placeholder values.
- Integrate a cost database lookup to assign realistic, component-specific costs based on market research.
- Add mandatory fields for material grades (e.g., ASTM standards) in the BOM generation pipeline.

Cost

Score: 8/10

EVALUATION REASONS

- Perfect unit-level arithmetic accuracy across the provided module values without orphaned balances.
- Hardware non-recurring engineering (NRE) costs are entirely unrealistic given the system's clear complexity and unit cost.
- Suspicious uniform pricing on auxiliary modules indicates a potential flaw in the depth of the estimation model.
- Arithmetic correctness is flawless, ensuring data reliability.
- Per-module breakdown effectively highlights cost drivers, but the absence of a ceiling and lack of NRE justification reduce overall rigor.

IMPROVEMENT SUGGESTIONS

- Add a heuristic check in the cost estimation pipeline to flag when more than two dissimilar sub-modules return the exact same integer cost, prompting dynamic recalculation.
- Introduce a baseline NRE scaling formula (e.g., minimum NRE = Unit Cost * Tooling/Engineering Complexity Multiplier) to prevent NRE estimates from falling below a single unit's BOM cost.
- Add automated sum validation in the report generation pipeline to catch arithmetic errors before output.
- Integrate a cost benchmarking module that requires ceiling or target cost input and provides industry comparisons for realistic range assessment.
- Include a target cost ceiling to enable overrun analysis and cost-performance comparisons.

Risks

Score: 7/10

EVALUATION REASONS

- Substantial content
- Contains numerical data
- Contains units of measurement

Suppliers

Score: 6/10

EVALUATION REASONS

- Contains numerical data
- Contains units of measurement

Research

Score: 6/10

EVALUATION REASONS

- Contains numerical data
- Contains units of measurement